



**Arab Academy for Science, Technology & Maritime Transport**  
**College of Artificial Intelligence ————— Alamein Campus**

|                    |                       |                        |                       |
|--------------------|-----------------------|------------------------|-----------------------|
| Course             | GN312                 | Embedded Systems & IoT | 2 <sup>nd</sup> 22/23 |
| Lecturer           | Dr. Omar Shalash      |                        |                       |
| Teaching Assistant | Hossam Eldeen Elsayed |                        |                       |

### Project Description:

In this project, you will design and implement an embedded system that utilizes Internet of Things (IoT) technology. The system should have at least two sensors and two actuators per person, and it should be able to communicate with other devices or services through the internet.

- The project must be approved by the course instructor.
- The deadline for the project proposal submission will be the 5<sup>th</sup> week.
- The deadline for the final project submission will be the 13<sup>th</sup> week.

### Requirements:

- **Hardware Design:** The project must include a custom hardware design that includes the sensors and the actuators. The hardware design must be submitted as a schematic or a PCB layout.
- **Software Development:** The project must include software development for the embedded system, which includes programming the microcontroller to read data from the sensor, process the data, and control the actuator. The software should also include networking capabilities to communicate with other devices or services through the internet.
- **IoT Integration:** The embedded system must be able to connect to an IoT platform or service (e.g., AWS IoT, Azure IoT, Google Cloud IoT or Similar platforms) and send data to the platform. The project must also include a web or mobile app that can display the data in real-time.
- **Documentation:** The project must include detailed documentation of the hardware and software design, including a block diagram, schematic, code documentation, and user manual.

### Grading Rubric:

| Criteria             | Points |
|----------------------|--------|
| Hardware Design      | 20     |
| Software Development | 30     |
| IoT Integration      | 30     |
| Documentation        | 20     |
| Total                | 100    |



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**Hardware Design (20 points):**

- Custom hardware design with at least two sensors and two actuators per person (10 points)
- PCB layout or schematic provided (5 points)
- Design meets specifications and is functional (5 points)

**Software Development (30 points):**

- Software developed to read data from the sensor, process the data, and control the actuator (15 points)
- Networking capabilities to communicate with other devices or services through the internet (10 points)
- Code is well-documented and organized (5 points)

**IoT Integration (30 points):**

- Embedded system connects to an IoT platform or service and sends data to the platform (15 points)
- Web or mobile app that can display the data in real-time (10 points)
- Integration is reliable and robust (5 points)

**Documentation (20 points):**

- Detailed documentation of the hardware and software design, including a block diagram, schematic, code documentation, and user manual (15 points)
- Documentation is clear and well-organized (5 points)

**Note: Any copied code from the internet or classmates will  
result in an instant 0**